# UMAP + KMEANS →

TSNE
umap

* Agenda → umap → graph

Y12

Unsupervised

kmeans → Viz, maths, code, evaluation

kmeans

* TSNE, PCA

Umap → 2020

* uniform manifold approx & projection

Dim Reduc

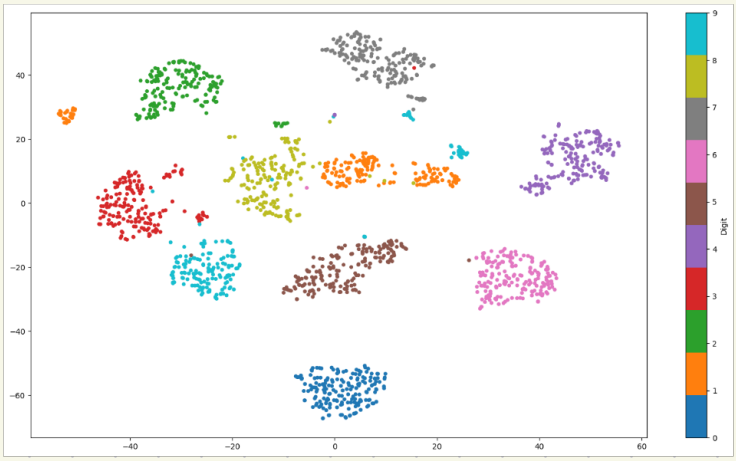
PCA, TSNE

High (5D,) → low (2D, 3D)

umap → local neighborhood
→ more global structure

fast

TSNE → very slow
clusters → meaningful




k

TSNE

Clustering

umap → faster

great

More global info

TSE A B

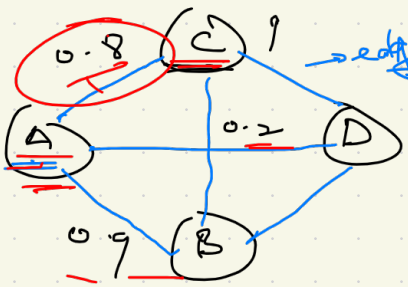
$x_i$

(1) Some hneighbors → Hyper params

(2) Builds a graph ($x_i$)

Comb$^n$ of edges & nodes

wt.




⇒

Google maps

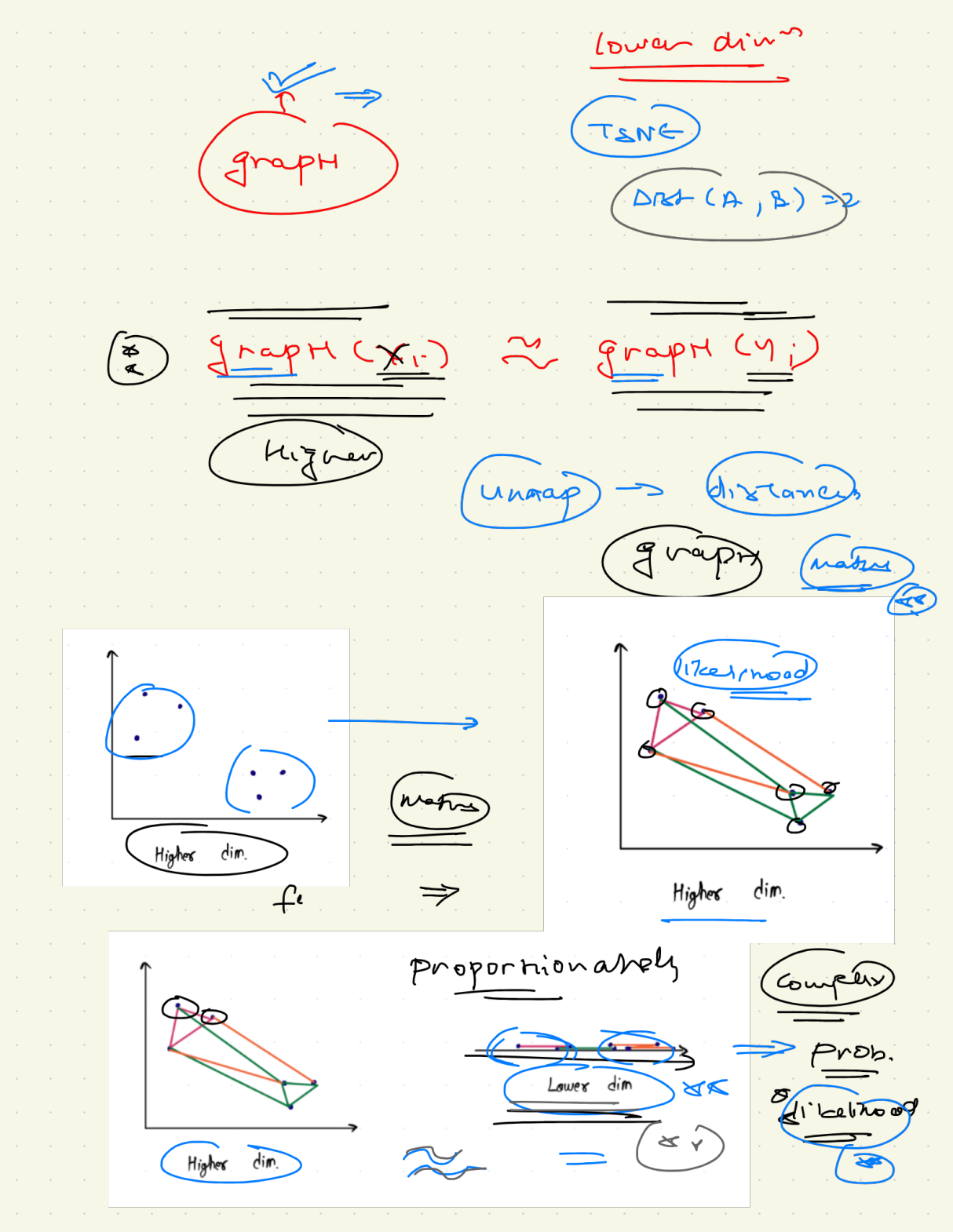
Traffic

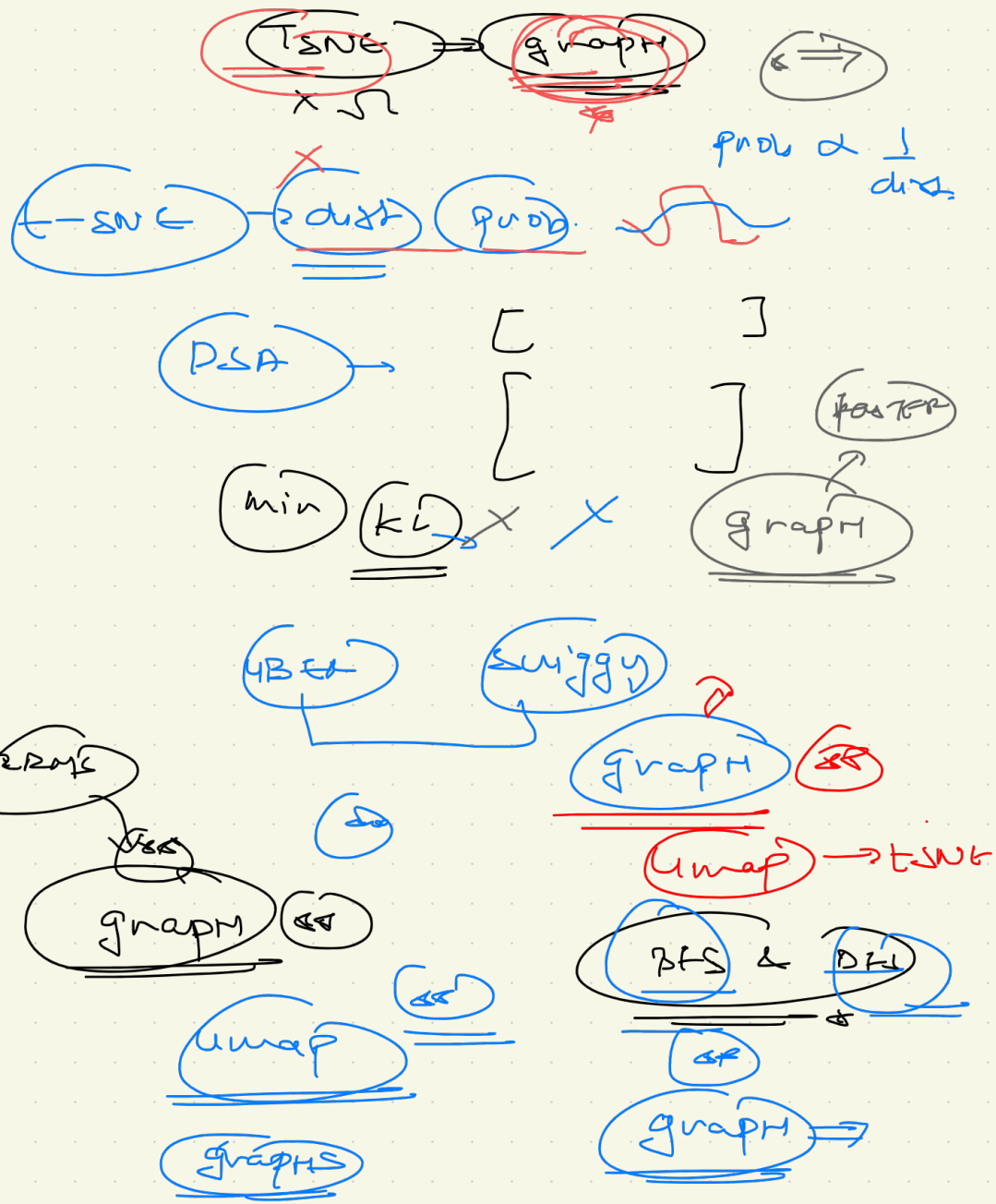
look point → find some neighbor

↓

(Builds a graph)

High dim


lower dim
graph
TSNE
DRT (A, B) =
graph (Xi)
graph (Yi)
Higher
Umap
distances
graph
matrix
likelihood
Higher dim.
matrix
fe
Higher dim.
proportionately
Lower dim
Complex
Prob.
likelihood


TSNE
graph
prob ∝ 1/dist
t-SNE
dist
prob
PCA
min
KL
graph
UMAP
graph
graph
UMAP
graphs
graph
UMAP
t-SNE
BFS & DFS
graph

Dimn Redn

Supervised → labelled data

$(X, Y)$

Regression → $y_i$ exists

Classification

regression

$$D = \{ (x_i, y_i)_{i=1}^{N} ; x_i \in R^d ; y_i \in R \}$$

$y_i \in \{0, 1\}$

$0, 1, 2$

Ex

Unsupervised → No o/p label

challenging

$$D = \{ (x_i)_{i=1}^{N} ; x_i \in R^d \}$$

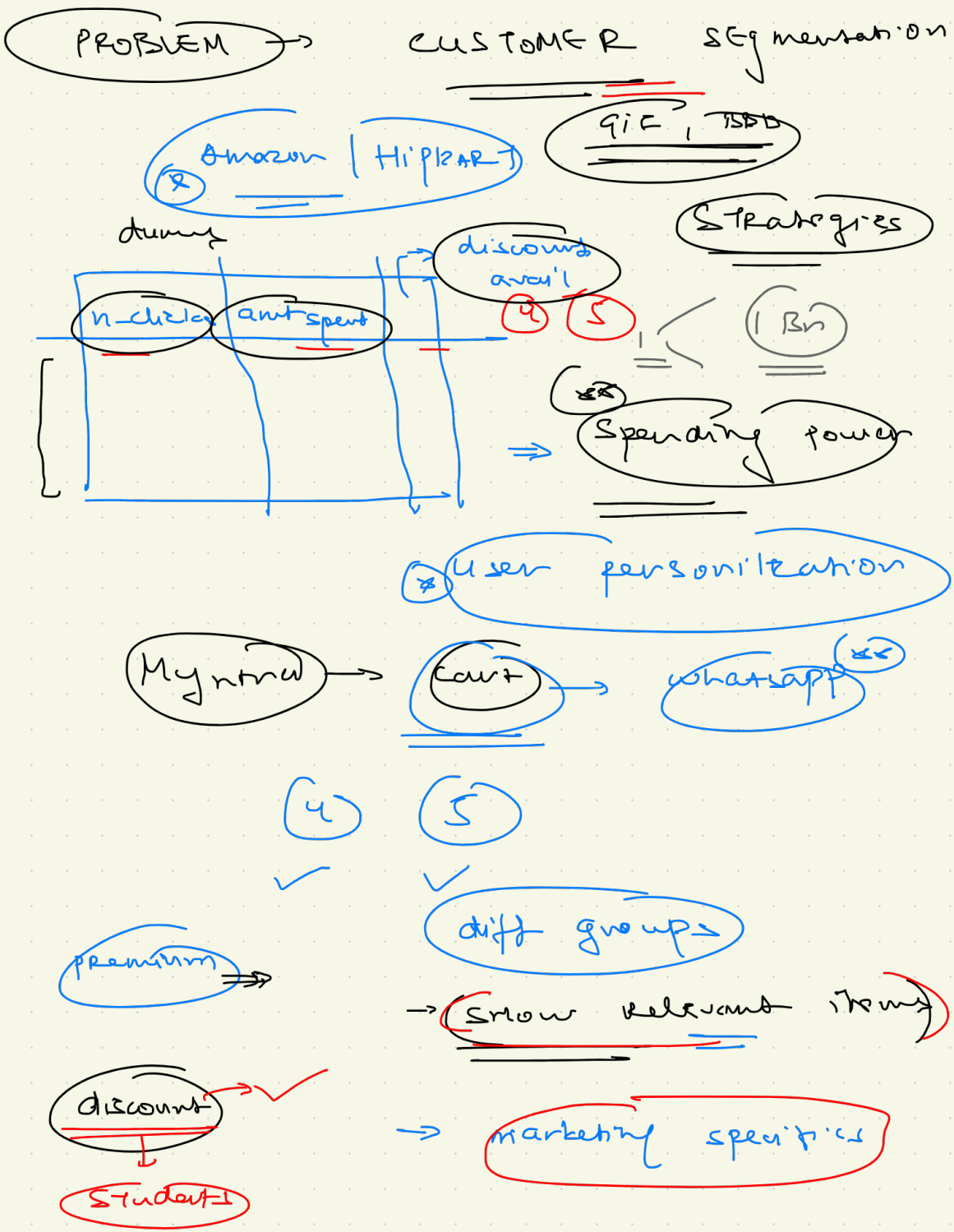
→ No $y_i$

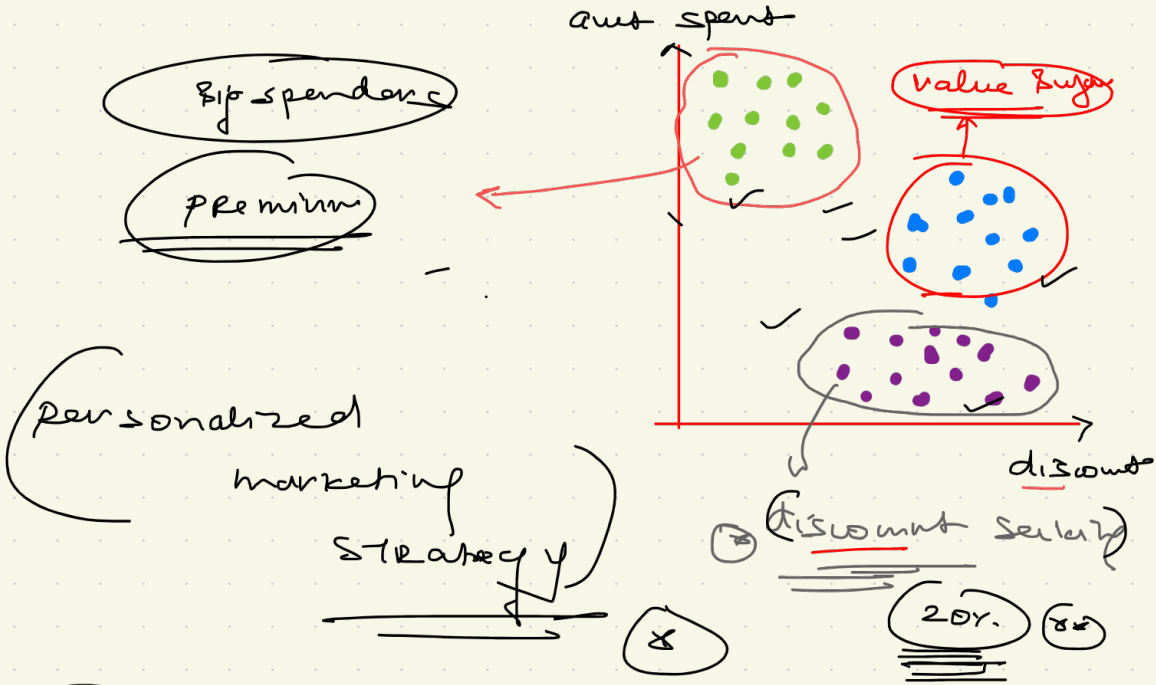
Supervised

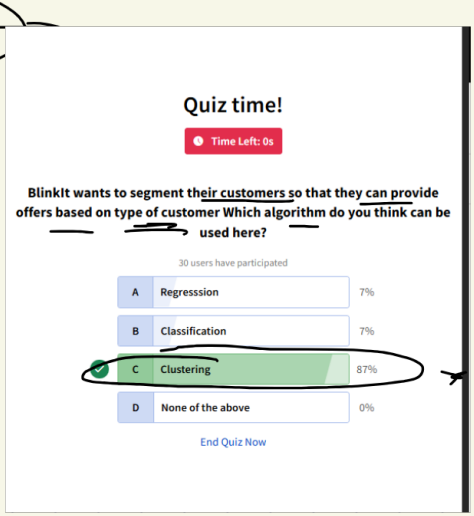
correct ness

[ Challenges → unavailability of labels

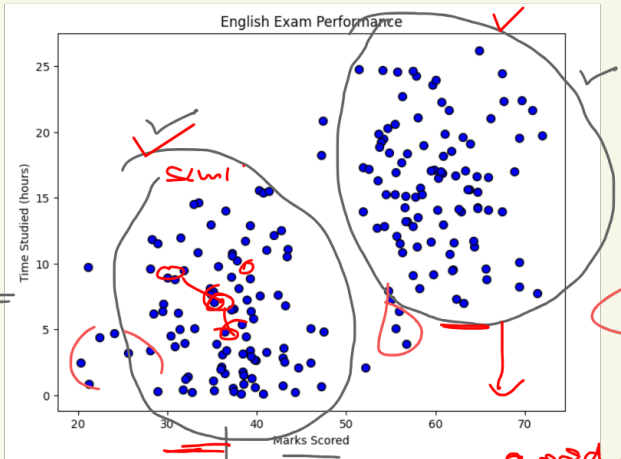
measure correctness ]


PROBLEM
CUSTOMER SEGMENTATION
GIF, JSON
Amazon
Hi Bazar
STRATEGIES
dummy
discount avail
n_clicks
amt spent
Spending power
User personilization
Mynthra
Cart
whatsapp
4
5
diff groups
premium
show relevant items
discount
Student
marketing specifics


amt spent
Big spenders
premium
Value Buyers
discount
personalized
marketing
strategy
discount seekers
20%.
idea


Quiz time!
Time Left: 0s
BlinkIt wants to segment their customers so that they can provide offers based on type of customer Which algorithm do you think can be used here?
30 users have participated
A Regresssion 7%
B Classification 7%
C Clustering 87%
D None of the above 0%
End Quiz Now

# Clustering intuition




Clusters

Outliers → clipping

Outliers

good marks

Studied less

DP

DP → should be similar

DP → ✗

Dissimilar students

metrics

cosine

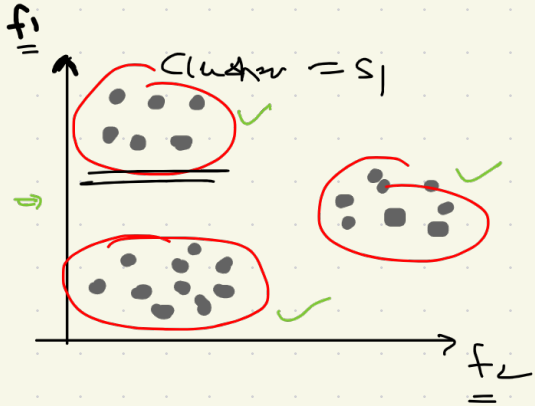
distance → $L_1$

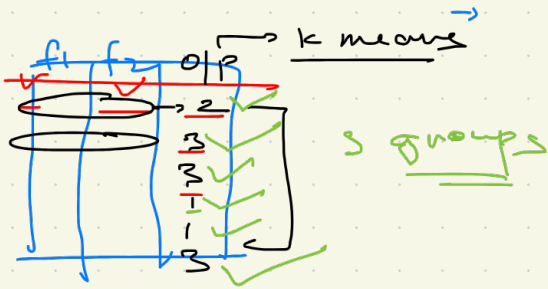
$L_2$

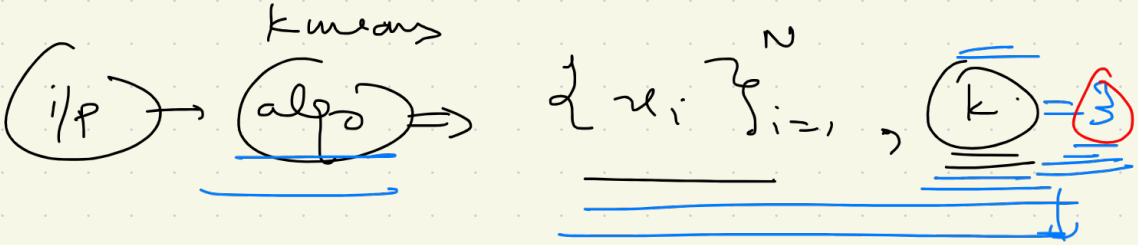
If k is 20 → High diff

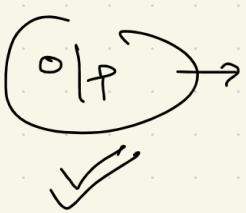
Algo → K Means → mean | avg

Hyperparameter → k ≃ No. of CLUSTERS



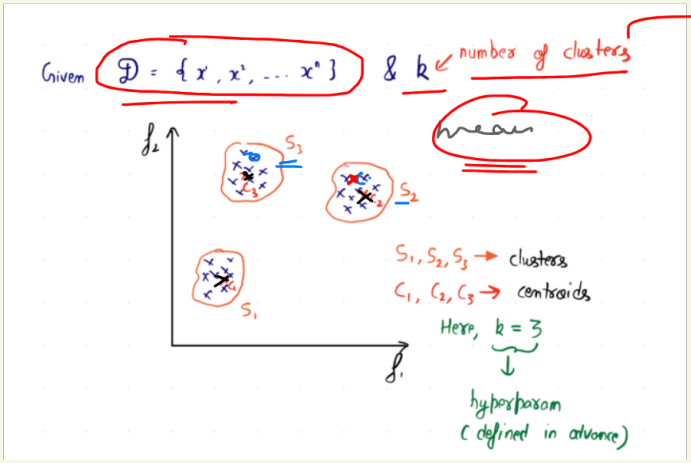




$S_i$ → set of pts belonging to a particular cluster

$S_1$ $S_2$ $S_3$

10:06

Break → 5 mins




hyperparam

k

$S_1$ $S_2$ $S_3$

$C_1$ $C_2$ $C_3$

centroid

k

k centroids

algo → data & k → CLUSTERS

Kmeans Algorithm

3

1 Initialization

3

Pick k points randomly

k

initial centroid

(2) Assignments

$x_i$ → D

(k) centroids

Dist.

pick nearest centroid

for each $x_i$ in D:

→ select nearest centroid → $c_j$

→ add to $S_j$

(3) Update CENTROIDS

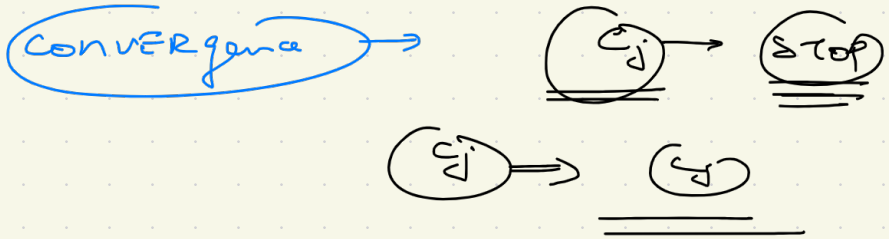
DP

for j in 1 to k:

$$c_j = \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$$

(2) & (3) → 100 ✓ 200

Convergence




(3) → (1) init randomly k centroids

(2) Assignments

(3) update centroids

n times or CONVERGence

Stop (1) no significants change in centroid

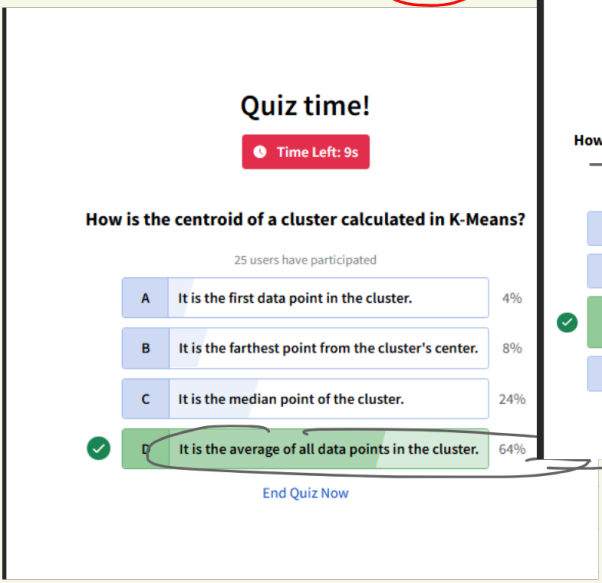
(2) u can run for fixed iterations

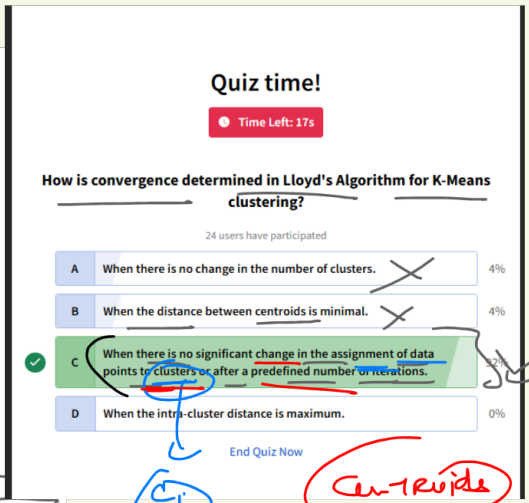
$S_2 = \{x_1, x_2, x_9, x_{21}, x_{17}\}$

$$c_2 = \frac{1}{|S_2|} \sum_{i=1}^{S_2} x_i$$

General

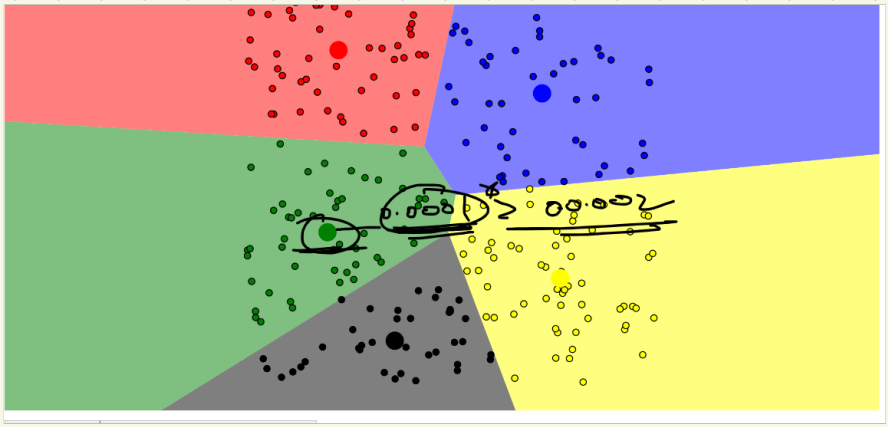
$$c_j = \frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$$



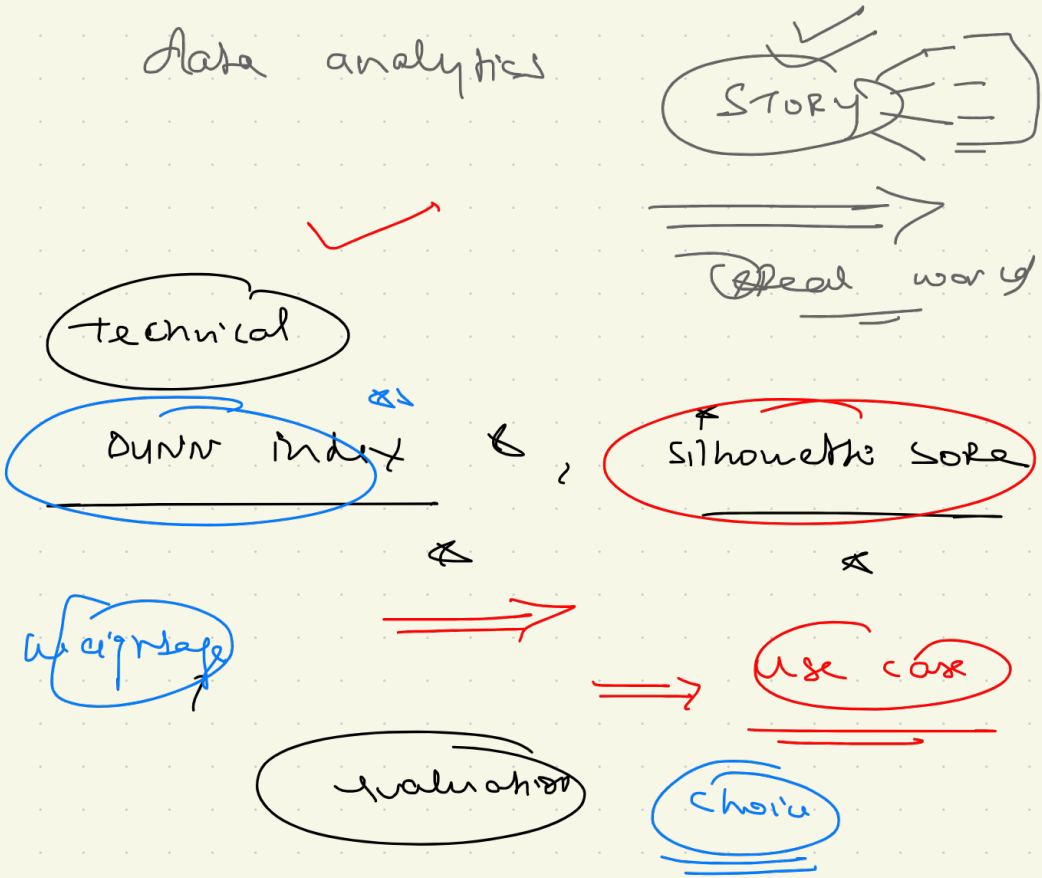

Centroids

DP



```
df.groupby('kmeans').mean()
```

| kmeans | ID | n_clicks | n_visits | amount_spent | amount_discount | days_since_registration | profile_information |
|---|---|---|---|---|---|---|---|
| 0 | 328.801158 | 250.683398 | 67.837838 | 3368.174213 | 83.436026 | 194.457529 | 214.345560 |
| 1 | 1205.137006 | 392.156780 | 88.338983 | 703.410958 | 154.006259 | 122.788136 | 198.072034 |
| 2 | 2114.945425 | 592.556982 | 127.125201 | 1430.665921 | 1144.668568 | 201.510433 | 197.481541 |
| 3 | 1202.124424 | 376.399386 | 91.099846 | 735.319192 | 162.652507 | 290.675883 | 197.087558 |


Data analytics
STORY
Real world
technical
DUNN Index
&
,
Silhouette Score
&
&
&
Usecase
Evaluation
Choice